# *B. Tech Degree IV Semester Examination April 2012*

## ME 404 APPLIED THERMODYNAMICS

(2006 Scheme)

Time : 3 Hours    Maximum Marks : 100

### PART A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) State and explain Carnot's Theorem.

(b) Define irreversibility and availability.

(c) Explain P-V-T surface.

(d) What is binary vapour cycle?

(e) Explain volumetric and gravimetric analysis of gas mixtures.

(f) Explain gas constant and specific heat of the gas mixtures.

(g) What are the various advantages of liquid and gaseous fuels over solid fuels?

(h) Explain adiabatic flame temperature.

### PART B

(4 x 15 = 60)

II. The internal energy of a certain substance is given by the equation $u = 3.56\ pv + 84KJ/Kg$, where u is in KJ/Kg, p is in Kpa and v is in $M^3/Kg$. A system consists of 3Kg of substance which expands from an initial pressure of 500KPa, volume of $0.22m^3$ to a final pressure of 100KPa in a process in which pressure and volume are given by the relation of $PV^{1.2} = C$. If the expansion is quasi static find the heat transfer, internal energy and work transfer for the process. In another process the same system expands according to the same pressure volume relationship and from the same initial state to final state but the heat transfer in this case is 30 KJ. Find the work transfer for the process. (15)

**OR**

III (a) Explain the Claussius Clapeyron equation. (7)

(b) Air flows steadily at the rate of 0.5 Kg/s through air compressor entering at 7m/s velocity, 100 KPa pressure, 0.95 $m^3/Kg$. volume and leaving at 5 m/s, 700 KPa and 0.19 $m^3/Kg$. The internal energy of the air leaving is 90 KJ/Kg greater than that of the air entering. Cooling water in the compressor jacket absorbs heat from the air at the rate of 58KW. Compute: (i) The rate of shaft work input (ii) The ratio of inlet pipe diameter to outlet pipe diameter. (8)

IV (a) Describe Rankine Cycle and explain how actual vapour cycle is different from Rankine Cycle. (7)

(b) Find the enthalpy and entropy of the steam when pressure is 2 Mpa and the specific volume is $0.09m^3/kg$. (8)

**OR**

V. (a) Explain any one of modern steam generator. (6)

(b) A Lancashire boiler generates 2400 KJ of dry steam per hour at a pressure of 11 bar. The grate area is $3m^2$ and 90kg of coal is burnt per $m^3$ of grate area per hour. The calorific value of coal is 33180 KJ/Kg and temperature of feed water is 17.5°C. Determine: (9)

(i) Actual evaporation per Kg. of coal.

(ii) Equivalent evaporation from and at 100°C

(iii) Efficiency of the boiler.

(P.T.O)

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| VI. | (a) | State and explain Gibb's -Dalton's law. | (6) |
| | (b) | A mixture consisting of 1 Kg of Helium and 2.5 Kg of Nitrogen at 25°C and 10 $Kg/cm^2$ is compressed in a reversible adiabatic process to 70 $Kg/cm^2$. Calculate: (i) partial pressure of the constituents (ii) final temperature (iii) change in internal energy of the mixture. Assume: | (9) |

| | Nitrogen | Helium |
|---|---|---|
| $C_V$ | 0.743 KJ/Kgk | 3.14KJ/Kgk |
| Cp | 1.04 KJ/Kgk | 5.23 KJ/Kgk |

**OR**

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| VII. | (a) | Explain the working of single stage reciprocating air compressor and obtain the work done in a polytropic compression without clearance volume. | (10) |
| | (b) | Explain Roots blower with neat figure. | (5) |
| VIII | (a) | Explain the bomb calorimeter. | (7) |
| | (b) | The products of combustion of hydro carbon fuel of unknown composition have the following composition as measured on a dry basis:<br>$CO_2$ 8% CO 0.9% $O_2$ 8.8% $N_2$ 82.3%<br>Calculate: (i) the air fuel ratio (ii) composition of the fuel on a mass basis (iii) percentage theoretical air on a mass basis. | (8) |

**OR**

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| IX. | (a) | Explain the application of first law of thermodynamics to combustion. | (6) |
| | (b) | A small gas turbine uses $C_8H_{18}$ ($\ell$) for fuel and 400 percent theoretical air. The air and fuel enter at 25°C and the products of combustion leave at 900K. The output of the engine and fuel consumption are measured and it is found that the specific fuel consumption is 0.25 Kg/s of fuel per MW output. Determine the heat transfer from the engine per Kmol of fuel. Assume complete combustion. | (9) |

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